

COSC 594/650

Optimization for Machine Learning

(Artificial Intelligence
Machine Decision-Making)

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This course focuses on optimization for machine learning
(**Selective** optimization problems, algorithms, and analyses)

Disclaimer

This is NOT

a classical optimization course

a machine learning course

a deep learning course

a decision-making course

Outline

- UNCONSTRAINED OPTIMIZATION
- CONSTRAINED OPTIMIZATION
- MINIMAX OPTIMIZATION
- STOCHASTIC OPTIMIZATION
- NONSMOOTH OPTIMIZATION
- NONCONVEX OPTIMIZATION
- OTHER TOPICS

(**ML examples, algorithms, and analyses**)

Logistics

- 5% CLASS PARTICIPATION
- 10% LECTURE SCRIBING ($\approx$ 1 WEEK)
- 20% HOMEWORK (5 ASSIGNMENTS, EACH 4%)
- 20% IN-CLASS MIDTERM (CLOSED-BOOK EXAM)
- 45% PROJECT (PROJECT GUIDELINE)

Check syllabus for use of AI

Prerequisites

■ LINEAR ALGEBRA AND CALCULUS

Matrix, rank, eigenvalue, singular value, subspace, linear dependence, inner product, norms; limits, derivative, Taylor expansion, continuity, smoothness

■ PROBABILITY

Random variable, probability distributions, expectation, conditional expectation, conditional probability, regression, Bayes' rule, maximum likelihood estimation

■ PROGRAMMING

Matrix or vector operations, for loops, and figure plots in Python

Course materials

■ SLIDES

Course slides will be shared on Canvas.

■ NOTES

Lecture notes from students will be shared on Canvas.

■ HOMEWORKS AND PROJECTS

Homework assignments will be announced on Canvas. Project guideline will also be communicated through Canvas.

Project ideas

■ YOUR ADVISORS

You are encouraged to talk to your advisors on potential problems

■ CONFERENCES

Check NeurIPS, ICML, ICLR, AISTATS, L4DC, OPT Workshop, etc

■ PAPER LIST

TBD

Acknowledgement

■ 6.881 OPTIMIZATION FOR MACHINE LEARNING

<https://optml.mit.edu/teach/6881/>

■ CSCI 599: OPTIMIZATION FOR MACHINE LEARNING

<https://spkreddy.org/optmlspring2025.html>

■ STAT9910-303: LARGE-SCALE OPTIMIZATION FOR DATA SCIENCE

https://yuxinchen2020.github.io/large_scale_optimization/index.html

■ EE364A: CONVEX OPTIMIZATION

<https://web.stanford.edu/class/ee364a/>

<https://web.stanford.edu/class/ee364b/>

■ IS NUMERICAL OPTIMIZATION THEORY IRRELEVANT TO MACHINE LEARNING PRACTICE IN 2026?

<https://icml.cc/virtual/2026/75373>

Terminology

Decision variable: $x \in \mathbb{R}^n$

Objective function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$

Feasible region: $X \subseteq \mathbb{R}^n$

■ OPTIMIZATION PROBLEM

minimize $f(x)$
 $x \in \mathbb{R}^n$
subject to $x \in X$

GOAL: Find a **best** decision

Motivating examples

Regression

Ratings for Knoxville restaurants (1–5)

data points/samples

Restaurant	Maya	Ethan	Sofia	Liam	Chloe
Tomato Head	4	3	5	4	5
Stock & Barrel	5	5	4	4	3
⋮	⋮	⋮	⋮	⋮	⋮
Yassin's Falafel	5	4	5	3	4

features/independent variables

■ PREDICTION

Should Chloe pay a visit to A Dopo Pizza?

Restaurant	Maya	Ethan	Sofia	Liam	Chloe
A Dopo Pizza	2	5	3	4	?

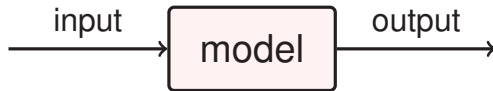
Learning problem

Input: Historical ratings other than Chloe

Output: Chloe's rating

Model: Mapping from input to output

■ PREDICTIVE MODEL



OBJECTIVE: Find a **best** model

Linear model

Ratings for Knoxville restaurants (1–5)

Input	Maya	Ethan	Sofia	Liam
x_1	4	3	5	4
x_2	5	5	4	4
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_m	5	4	5	3

$$y_i = \theta^\top x_i$$

Input	Chloe
y_1	5
y_2	3
$\vdots$	$\vdots$
y_m	4

■ LINEAR MODEL

$$y_i = \theta^\top x_i$$

Linear regression (I)

Ratings for Knoxville restaurants (1–5)

Input	Maya	Ethan	Sofia	Liam
x_1	4	3	5	4
x_2	5	5	4	4
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_m	5	4	5	3

$$y_i = \theta^\top x_i \longrightarrow$$

Input	Chloe
y_1	5
y_2	3
$\vdots$	$\vdots$
y_m	4

■ ℓ_2 REGRESSION

$$\begin{array}{ll} \underset{\theta}{\text{minimize}} & \sum_{i=1}^m \left(\theta^\top x_i - y_i \right)^2 \\ \text{subject to} & \theta \in \mathbb{R}^4 \end{array}$$

Linear regression (II)

Ratings for Knoxville restaurants (1–5)

Input	Maya	Ethan	Sofia	Liam
x_1	4	3	5	4
x_2	5	5	4	4
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_m	5	4	5	3

$$y_i = \theta^\top x_i$$

Input	Chloe
y_1	5
y_2	3
$\vdots$	$\vdots$
y_m	4

■ ℓ_1 REGRESSION

$$\begin{array}{ll} \underset{\theta}{\text{minimize}} & \sum_{i=1}^m \left| \theta^\top x_i - y_i \right| \\ \text{subject to} & \theta \in \mathbb{R}^4 \end{array}$$

Linear regression (III)

Ratings for Knoxville restaurants (1–5)

Input	Maya	Ethan	Sofia	Liam
x_1	4	3	5	4
x_2	5	5	4	4
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_m	5	4	5	3

$$y_i = \theta^\top x_i$$

$\xrightarrow{\hspace{1cm}}$

Input	Chloe
y_1	5
y_2	3
$\vdots$	$\vdots$
y_m	4

■ ℓ_∞ REGRESSION

$$\begin{array}{ll} \underset{\theta}{\text{minimize}} & \max_i \left| \theta^\top x_i - y_i \right| \\ \text{subject to} & \theta \in \mathbb{R}^4 \end{array}$$

Logistic regression

Ratings and binary recommendations for Knoxville restaurants

Restaurant	Maya	Ethan	Sofia	Liam	Chloe
Tomato Head	4	3	5	4	1
Stock & Barrel	5	5	4	4	0
⋮	⋮	⋮	⋮	⋮	⋮
Yassin's Falafel	5	4	5	3	1
A Dopo Pizza	2	5	3	4	?

$y_i = 1$: come

$y_i = 0$: not come

■ BINARY CLASSIFICATION

Would Chloe visit **A Dopo Pizza**?

Logistic model

Input	Maya	Ethan	Sofia	Liam
x_1	4	3	5	4
x_2	5	5	4	4
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_m	5	4	5	3

$$p_i = \sigma(\theta^\top x_i)$$

$\xrightarrow{\hspace{1cm}}$

Input	Chloe
y_1	1
y_2	0
$\vdots$	$\vdots$
y_m	1

■ LOGISTIC MODEL

$$\mathbb{P}(y_i = 1 \mid x_i) = \sigma(\theta^\top x_i)$$

$$\mathbb{P}(y_i = 0 \mid x_i) = 1 - \sigma(\theta^\top x_i)$$

★ sigmoid function

$$\sigma(z) = \frac{1}{1 + \exp(-z)}$$

Logistic regression

Probability of observing y_i : $(p_i)^{y_i} (1 - p_i)^{1-y_i}$ for $y_i \in \{0, 1\}$

■ MAXIMUM LIKELIHOOD

$$\begin{array}{ll} \underset{\theta}{\text{maximize}} & \log \prod_{i=1}^m (p_i)^{y_i} (1 - p_i)^{1-y_i} \\ \text{subject to} & \theta \in \mathbb{R}^4 \end{array}$$

★ negative log-likelihood minimization

$$\begin{array}{ll} \underset{\theta}{\text{minimize}} & - \sum_{i=1}^m [y_i \log \sigma(\theta^\top x_i) + (1 - y_i) \log(1 - \sigma(\theta^\top x_i))] \\ \text{subject to} & \theta \in \mathbb{R}^4 \end{array}$$

Optimization in machine learning

■ LINEAR REGRESSION

$$\begin{array}{ll} \underset{\theta}{\text{minimize}} & \text{loss function} \left(\{\theta^\top x_i, y_i\}_{i=1}^m \right) \\ \text{subject to} & \theta \in \mathbb{R}^d \end{array}$$

- ★ $\ell_2 / \ell_1 / \ell_\infty$ loss functions
- ★ logistic loss function

QUESTIONS: How to choose an objective? What are useful properties?
How should we optimize it?

NEXT: Review linear algebra, probability, and mathematical analysis

**Any suggestions on lectures and projects
are welcome.**

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